

C8 — The Periodic Table — Cheat Sheet

Key Terms

Periodic Table	A table of elements arranged in order of increasing atomic number, with elements of similar properties in vertical columns (groups)
Group	A vertical column in the Periodic Table; elements in the same group have the same number of outer electrons and similar chemical properties
Period	A horizontal row in the Periodic Table; the period number equals the number of electron shells
Alkali metals	The highly reactive metals in Group I (Li, Na, K, Rb, Cs); they react with water to form alkaline solutions
Halogens	The reactive non-metals in Group VII (F, Cl, Br, I); they exist as diatomic molecules
Halides	Compounds formed by halogens with metals or hydrogen (e.g. NaCl, HBr, KI)
Noble gases	The very unreactive elements in Group VIII (He, Ne, Ar, Kr, Xe, Rn); they have complete outer electron shells
Transition elements	The block of metals in the centre of the Periodic Table; they are hard, dense, have high melting points, form coloured compounds, and act as catalysts
Metalloid (semi-metal)	An element with properties between metals and non-metals; they are semiconductors (e.g. silicon, germanium)
Displacement reaction	A reaction where a more reactive element replaces a less reactive element in a compound
Diatomic molecule	A molecule consisting of two atoms bonded together (e.g. Cl ₂ , Br ₂ , I ₂ , H ₂ , O ₂)
Monatomic	Existing as single atoms (not bonded into molecules); noble gases are monatomic
Periodic property	A property that shows a repeating pattern across the periods of the Periodic Table
Electronic configuration	The arrangement of electrons in shells around the nucleus of an atom (e.g. 2,8,1 for sodium)
Catalyst	A substance that speeds up a chemical reaction without being used up itself
Valency	The combining power of an atom; halogens have a valency of 1

Key Equations

- > alkali metal + water → metal hydroxide + hydrogen
- > $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
- Physical state changes from gas → liquid → solid.
- > $\text{Cl}_2\text{(aq)} + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2\text{(aq)}$
- > $\text{Cl}_2\text{(aq)} + 2\text{KI(aq)} \rightarrow 2\text{KCl(aq)} + \text{I}_2\text{(aq)}$
- > $\text{Br}_2\text{(aq)} + 2\text{KI(aq)} \rightarrow 2\text{KBr(aq)} + \text{I}_2\text{(aq)}$
- 1. $2\text{Li(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{LiOH(aq)} + \text{H}_2\text{(g)}$
- 2. $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
- 3. $2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$
- 4. alkali metal + water → metal hydroxide + hydrogen
- 5. $\text{Cl}_2\text{(aq)} + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2\text{(aq)}$
- 6. $\text{Cl}_2\text{(aq)} + 2\text{KI(aq)} \rightarrow 2\text{KCl(aq)} + \text{I}_2\text{(aq)}$
- 7. $\text{Br}_2\text{(aq)} + 2\text{KI(aq)} \rightarrow 2\text{KBr(aq)} + \text{I}_2\text{(aq)}$
- Answer: lithium + water → lithium hydroxide + hydrogen
- Answer: Only chlorine (and fluorine) will displace bromine, because they are more reactive. Iodine will NOT displace bromine because it is less reactive. $\text{Cl}_2\text{(aq)} + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2\text{(aq)}$.
- Elements change from metallic → non-metallic across a period.
- Group I (alkali metals): soft, low density, low m.p., very reactive with water → $\text{MOH} + \text{H}_2$. Reactivity increases down the group.
- Group VII (halogens): diatomic (Cl₂, Br₂, I₂), poisonous, coloured. Reactivity decreases down the group. More reactive halogen displaces less reactive: $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$.

Exam Focus

- Group I trends and reactions: reactivity increases down the group; reaction with water produces metal hydroxide + hydrogen; linking trends to atomic

structure (outer electron further from nucleus).

- Group VII trends and displacement: reactivity decreases down the group; halogen displacement reactions with halide solutions; writing equations.
- Electronic configuration and position: using group number (= outer electrons) and period number (= shells) to predict or identify an element's position.
- Comparing Group I vs transition metals: listing differences in physical properties (hardness, density, melting point, reactivity) and distinctive transition metal properties (coloured compounds, variable oxidation states, catalysts).
- Noble gases: explaining unreactivity in terms of full outer electron shells; monatomic nature.
- Predicting properties: using trends within a group to predict melting points, reactivity, or physical state of elements not directly studied (e.g. rubidium, astatine).

Key Facts

- Major regions of the Periodic Table:
- Metals are found on the left and centre of the Periodic Table.
- Non-metals are grouped in the top right-hand region, above the thick stepped line.
- Metalloids have properties of both metals and non-metals:
- Group number = number of electrons in the outer shell (for main-group elements).
- Period number = number of electron shells the atom has.
- Why elements in the same group have similar properties:
- Across a period (left to right):
- Densities and melting points increase down any group (general trend).
- Trends down Group I (physical):
- Example — sodium with water:
- Trend: Chemical reactivity increases down the group:
- Why reactivity increases down the group:
- Common properties of all halogens:
- Trends down Group VII (physical):
- Why reactivity decreases down the group:
- Reactivity order: Chlorine > Bromine > Iodine
- Displacement summary table:
- Rule: A halogen can only displace a halogen below it in Group VII (a less reactive one).
- Hard and strong (not soft like alkali metals).
- High melting and boiling points (e.g. tungsten melts at 3410 °C — used in light bulb filaments).
- Iron — catalyst in the Haber process (making ammonia).
- Platinum and rhodium — catalysts in catalytic converters.
- Nickel — catalyst in hydrogenation of vegetable oils.
- Common transition metals and their uses:

Quick Q&A

Question	Answer
What is the name of the alkali formed when potassium reacts with water?	Potassium hydroxide (KOH).
Write a word equation for the reaction between lithium and water.	lithium + water → lithium hydroxide + hydrogen
Which halogen(s) will displace bromine from a solution of potassium bromide?	Only chlorine (and fluorine) will displace bromine, because they are more reactive. Iodine will NOT displace bromine because it is less reactive. Cl_2
The figure shows an outline of the Periodic Table with elements V, W, X, Y marked. V is in Group I Period 4, Y is in Group II Period 4, W is a transit	X is in Group VII (a non-metal/halogen) and Y is in Group II (a metal/alkaline earth). So the correct answer is: Non-metal = X, Metal = Y. V is also a
(a) State whether transition elements have high or low densities. (b) State whether transition elements are all gases or all metals. (c) Give the name	(a) Transition elements have high densities. (b) They are all metals. (c) The noble gases (Group VIII / Group 0).